24DS636 - STATISTICAL MODELLING

Credit Card Launch Strategy for Atliqo Bank

AADITH P

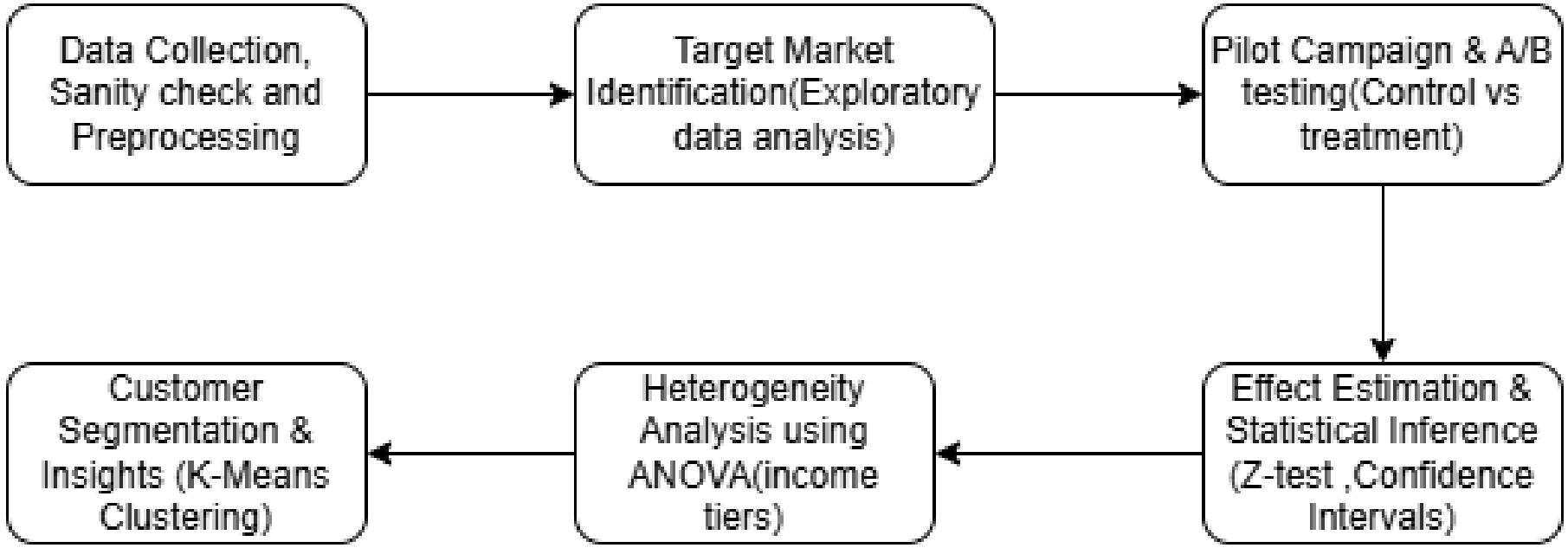
Introduction

- Credit cards are one of the fastest-growing retail financial products, yet most launch strategies rely on broad demographic targeting rather than data-driven identification of high-potential customer segments leading to inefficient acquisition spend and missed opportunities among underserved groups.
- Statistical modelling bridges this gap by combining Exploratory Data Analysis, hypothesis testing, and unsupervised learning into a structured pipeline that identifies the right target audience, validates campaign effectiveness with statistical techniques, and discovers actionable customer sub-segments.

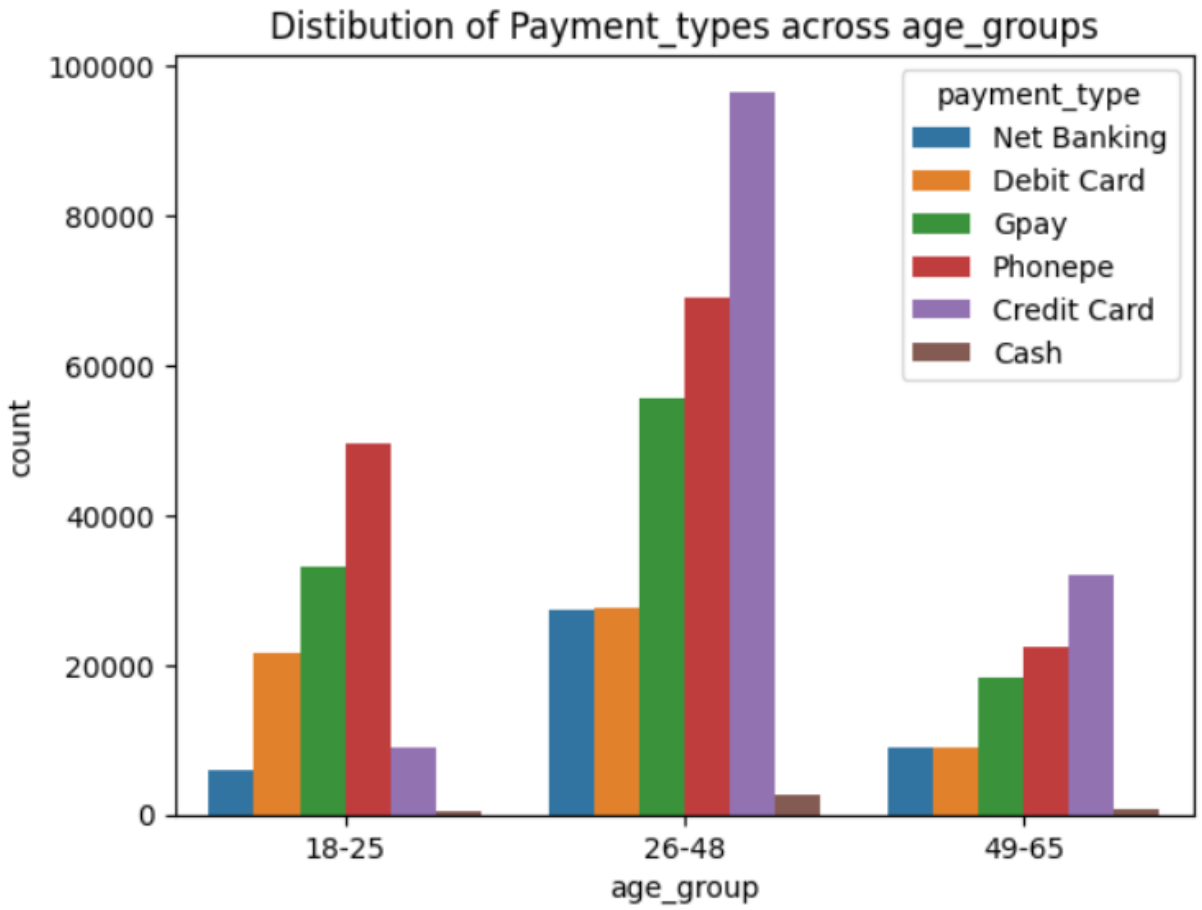
PROBLEM STATEMENT

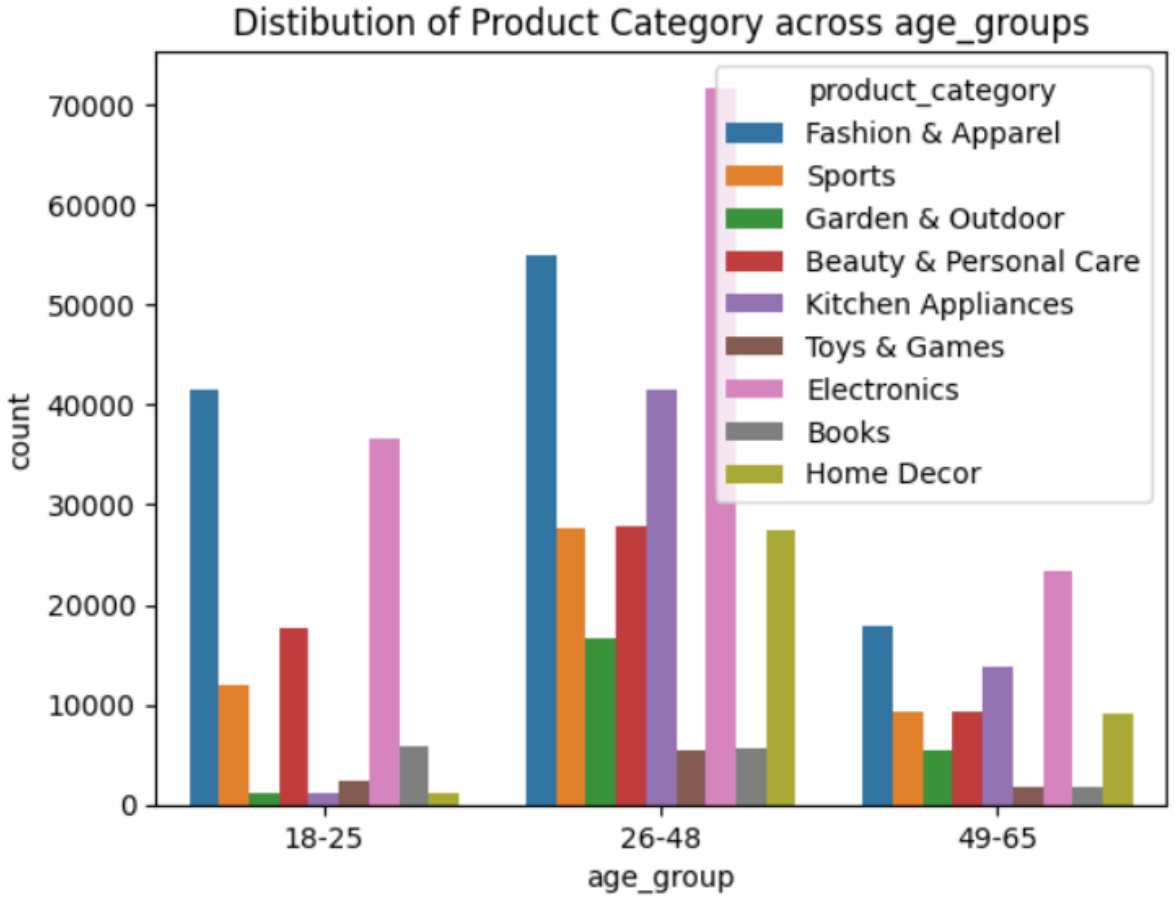
- Atliqo Bank is a newly established bank entering the Indian credit card market.
- The market is highly competitive, dominated by players such as HDFC Bank, ICICI Bank, and Axis Bank.
- Due to strong competition, targeting the right customer segment is critical.
- The challenge is to make these decisions using statistical evidence, not assumptions.
- A poorly targeted credit card launch can lead to high default risk, low card utilization, and increased operational losses, especially for a new entrant with limited historical customer data.

Methodology



Results(EDA)

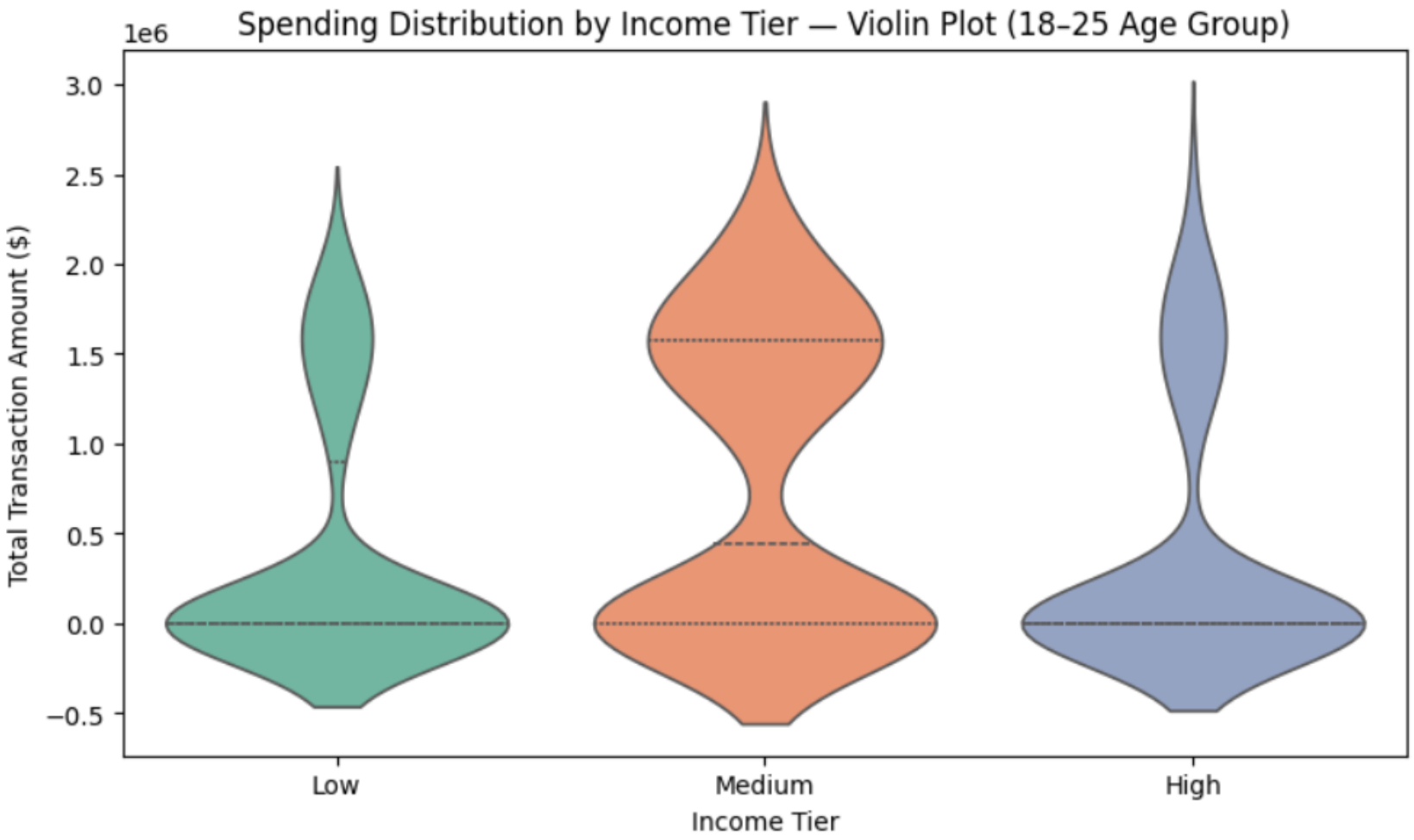

Distibution of Payment_types across age_groups
100000
80000
60000
40000
20000
0
count
18-25
26-48
49-65
age_group
payment_type
Net Banking
Debit Card
Gpay
Phonepe
Credit Card
Cash


Distibution of Product Category across age_groups
70000
60000
50000
40000
30000
20000
10000
0
count
18-25
26-48
49-65
age_group
product_category
Fashion & Apparel
Sports
Garden & Outdoor
Beauty & Personal Care
Kitchen Appliances
Toys & Games
Electronics
Books
Home Decor

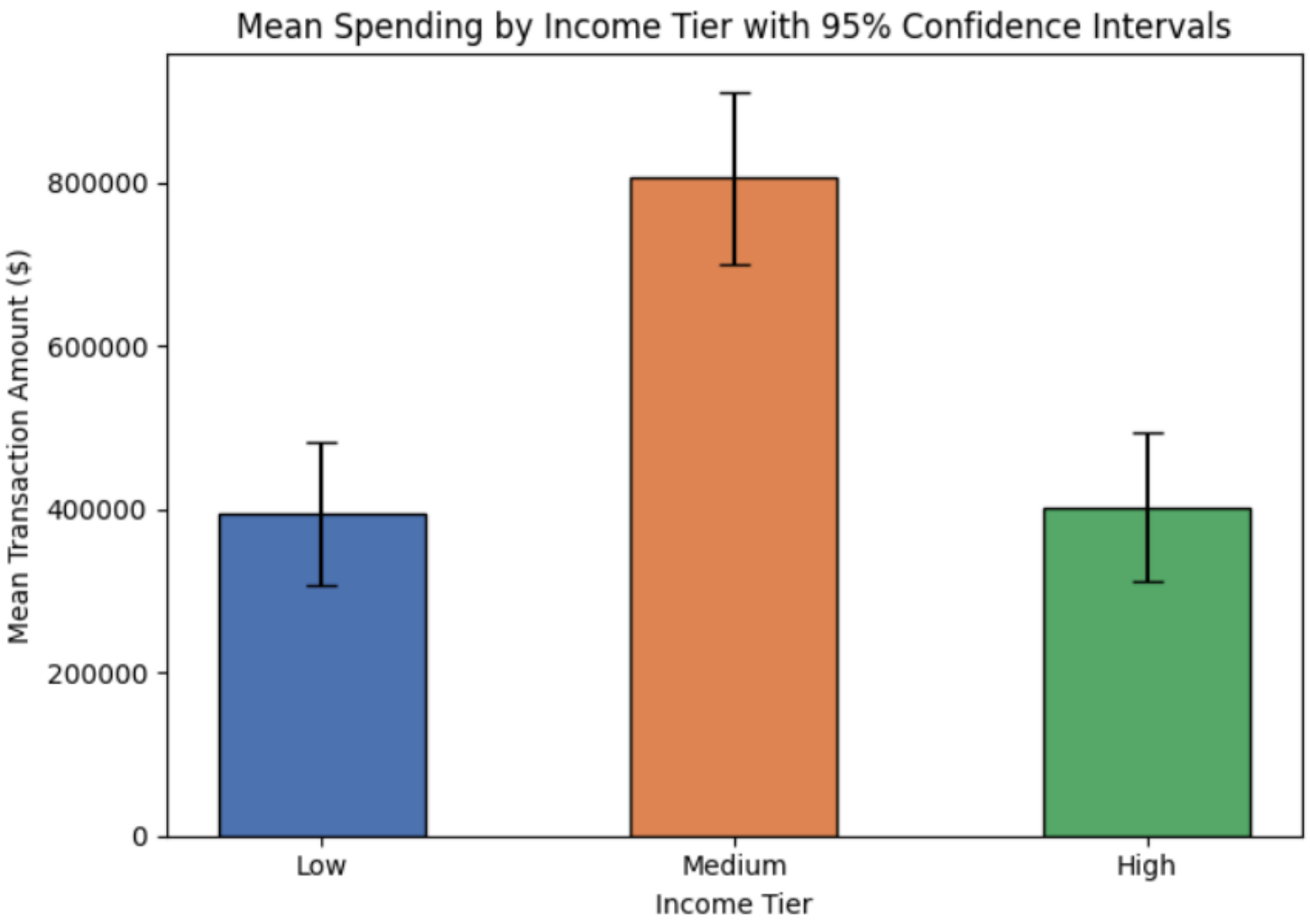
A/B Testing using Z-Test Inference

- **Hypothesis:** H_0 states the new credit card does not increase average transaction amount compared to the control group. H_1 states it does This is a right-tailed test targeting the 18–25 age group.
- **Test Result:** The computed Z-score exceeded the critical value of 1.645 ($\alpha = 0.05$, right-tailed), placing the observed statistic firmly inside the rejection region — H_0 is rejected.
- **Conclusion:** The test group (new card users) showed a statistically significant higher average transaction amount than the control group, validating that the credit card campaign produced a genuine behaviour lift among the 18–25 target segment.

Results(Heterogeneity Analysis using Anova)



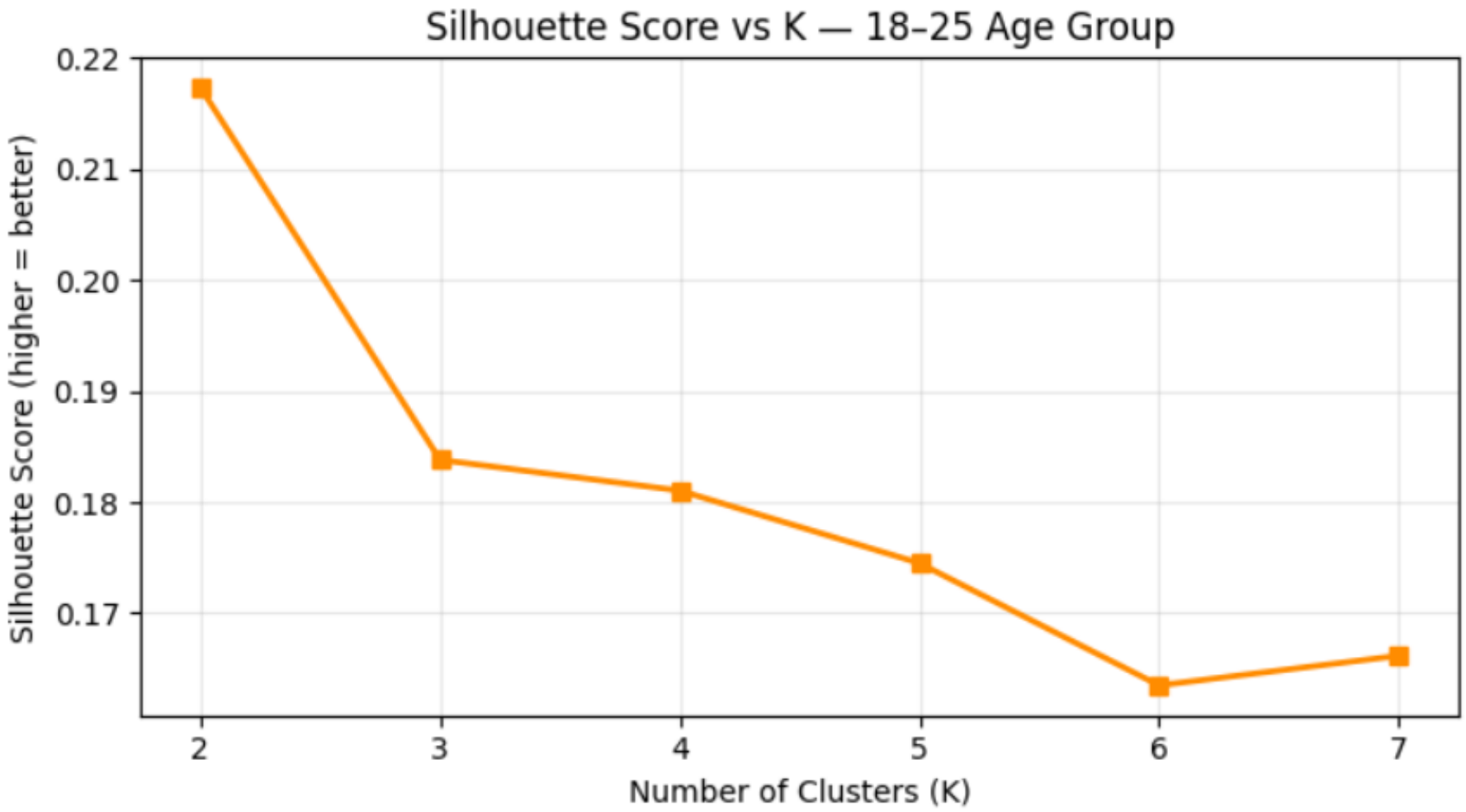
Mean spending bar chart with 95% CI error bars



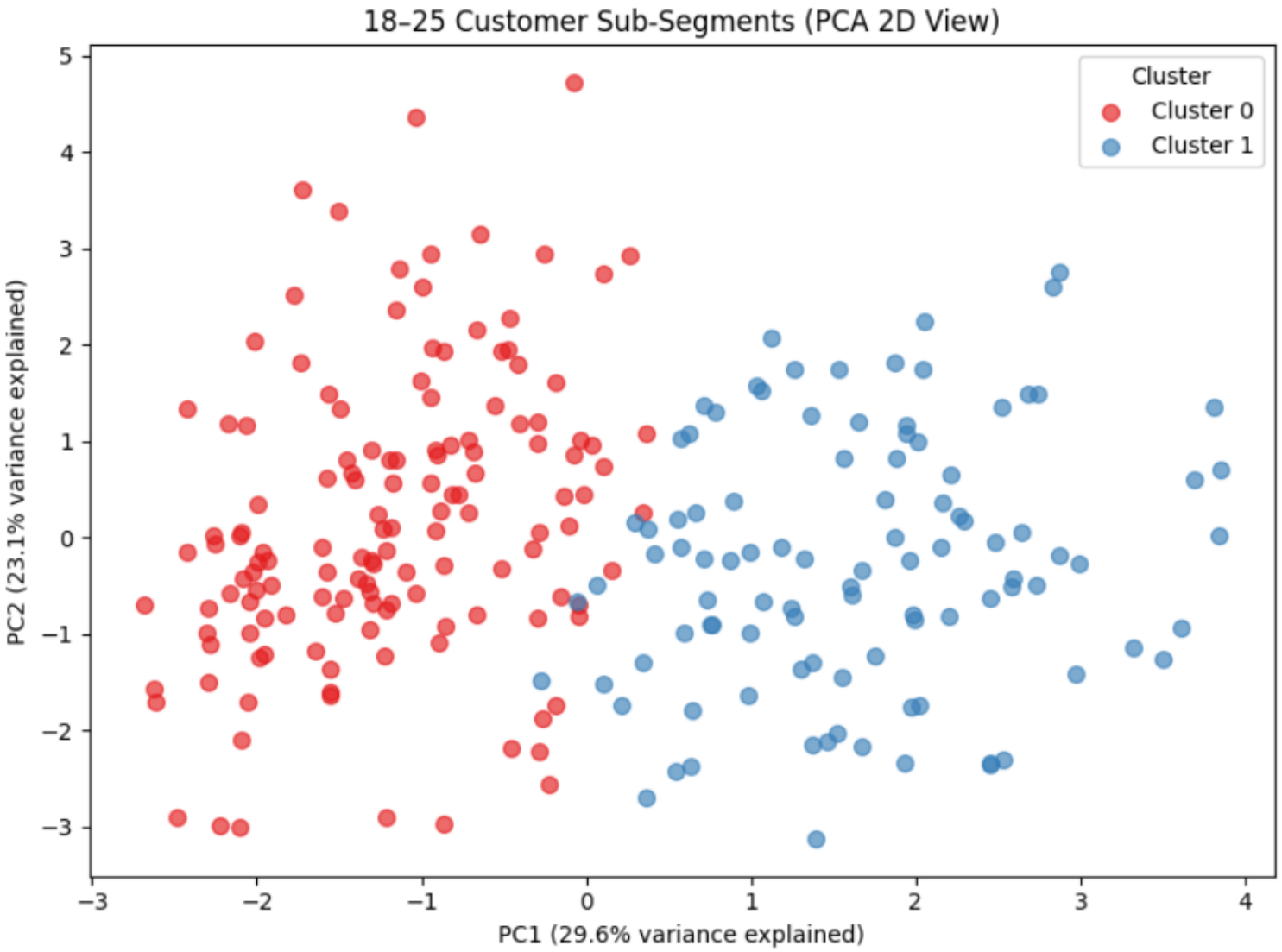
Tukey HSD Table

Group 1	Group 2	Mean Diff	p-adj	Lower	Upper	Reject
High	Low	-8,490.48	0.9916	-1,70,099.99	1,53,119.04	False
High	Medium	4,03,492.64	0	2,41,883.12	5,65,102.16	True
Low	Medium	4,11,983.12	0	2,50,373.60	5,73,592.64	True

Silhouette score plot



Cluster Separation Plot



CUSTOMER SEGMENT PROFILES — 18 to 25 Age Group

- Cluster 0 (140 customers)
  - Dominant Income Tier : Medium
  - Income Tier Breakdown : Low=22.1% | Medium=49.3% | High=28.6%
  - Avg Credit Score : 407
  - Credit Utilisation : 0.48
  - Avg Total Spending : $1,589,896
  - Avg Txn Amount : $3,197
  - Avg Txn Count : 497

CUSTOMER SEGMENT PROFILES — 18 to 25 Age Group

- Cluster 1 (100 customers)
  - Dominant Income Tier : Medium
  - Income Tier Breakdown : Low=30.0% | Medium=51.0% | High=19.0%
  - Avg Credit Score : 587
  - Credit Utilisation : 0.56
  - Avg Total Spending : $1,620,345
  - Avg Txn Amount : $3,239
  - Avg Txn Count : 500

Conclusion

- The credit card campaign is statistically effective for the 18–25 segment (A/B testing).
- Income significantly influences spending behavior (ANOVA results).
- Medium income group shows the highest spending potential (Tukey analysis).
- Customer behavior is heterogeneous, not uniform across segments.
- Cluster 1 (high creditworthiness + high activity) identified as the most valuable segment.

THANK YOU